

Xingyue Huang

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Education

University of Oxford 2023 – 2026
DPhil in Computer Science

Advisors: Prof. Michael Bronstein (DeepMind Chair of AI), Prof. Ismail Ceylan

University of Oxford 2019 – 2023
MMathCompSci in Mathematics and Computer Science
Graduated with Distinction

Selected Publications

[Threshold Differential Attention for Sink-Free, Ultra-Sparse, Non-Dispersive LLMs](#) — **ACL 2026** (First Author)
[Hierarchical Token Prepending for Long-Context LLM Embeddings](#) — **ACL 2026** (Co-First Author)
[HYPER: A Foundation Model for Link Prediction with Knowledge Hypergraphs](#) — **ICLR 2026** (First Author)
[Flock: A Knowledge Graph Foundation Model via Learning on Random Walks](#) — **ICLR 2026** (Co-First Author)
[How Expressive are Knowledge Graph Foundation Models?](#) — **ICML 2025** (First Author)
[Link Prediction with Relational Hypergraphs](#) — **TMLR** (First Author)
[Cooperative Graph Neural Networks](#) — **ICML 2024**
[A Theory of Link Prediction via Relational Weisfeiler-Leman on Knowledge Graph](#) — **NeurIPS 2023** (First Author)

Professional Experience

Snap Inc. - Research Intern (UMaP) 06/2025 - 10/2025

- Developed Threshold Differential Attention (TDA), a sink-free, ultra-sparse attention mechanism for long-context LLMs, achieving > 99% exact zeros while matching Softmax accuracy on QA benchmarks.
- Demonstrated long-context robustness on the SCROLLS benchmark and passkey-retrieval stress tests, where TDA outperformed Softmax by $\sim 2.5\times$ at 4k-token contexts.
- Co-authored *Hierarchical Token Prepending*, improving long-document embeddings in decoder-only LLMs via block-level summary tokens, with 5% gains across 11 retrieval datasets and 30 embedding benchmarks.

Eigent-AI - Research Intern 10/2024 – 06/2025

- Built a tool-use synthetic data generation pipeline producing 20k verified execution traces for CAMEL-AI.
- Used back-translated tool trajectories for supervised fine-tuning, improving math benchmark accuracy by 5%.
- Led the *Loong* verifier-driven RL framework for long chain-of-thought synthesis.

Alibaba Group - Machine Learning Engineer Intern 07/2021 – 09/2021

- Developed an object detection system for video subtitle-detection with Faster-RCNN model
- Improved accuracy of object detection and classification by 10% and were incorporated into production

Technical Skills

Systems & Infrastructure: Triton, LLM architecture design, multi-agent systems, post-training

ML & Data: PyTorch, HuggingFace, CAMEL-AI, SQL, DuckDB, pandas, torch-geometric

Services

Reviewer: NeurIPS 2025 **Top Reviewer** · ICML 2026 **Silver Reviewer** · ICLR 2026

Workshop Organizer: NeurIPS 2025 SEA, ICML 2026 GFM

Talks: Tutorial on Graph Foundation Models (LoG 2025) · Relational Hypergraphs (TU Wien Guest Lecture on GRL 2026/03/19, LoG 2024 Oxford Meetup)